Experiment16: Feed Forward Neural Network (FNN)

Aim

To write a Python program to implement a simple Feed Forward Neural Network.

Procedure

1. Read two input values from the user.
2. Assign weights and bias.
3. Calculate the weighted sum.
4. Apply the sigmoid activation function.
5. Display the output of the neural network.

Source Code

main.py Visualize Run

```
import math

x1 = float(input("Enter first input: "))
x2 = float(input("Enter second input: "))

w1 = 0.5
w2 = 0.4
b = 0.2

net = x1 * w1 + x2 * w2 + b

output = 1 / (1 + math.exp(-net))

print("Output of Feed Forward Neural Network =", round(output, 4))
```

Output

```
Enter first input: 2
Enter second input: 3
Output of Feed Forward Neural Network = 0.9168
```

Result

Thus, the Feed Forward Neural Network was implemented successfully.